

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 2

Year: 2017

Subject: 341152 Electric Circuits II

Section: 5-6

Date: 25 May 2018

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Instructions:

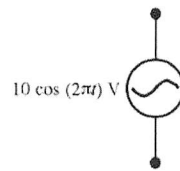
1. The examination has 12 pages (including this page, 3 parts (50 questions) and a total score of 90 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. In part 1 and 2, enter your answer in the answer sheet on the **LAST PAGE**.
11. In part 3, show all the steps of your solution with answers in the provided space underneath of each problem.

Cheating in the exam is considered an extremely serious offence
which will result in expulsion from the University.

Part 1 (25 points) True/False

Instruction: Following questions, mark "X" for the correct answer, "True" or "False" in the provided answer sheet (see LAST PAGE).

1. The voltage of independent source is changed by changing the element value in the circuit.
2. The Norton equivalent circuit can be converted to Thevenin equivalent circuit by using an AC source conversion method.
3. The voltage of dependent source is controlled by either a current or a voltage in circuit.
4. There is no unit of power factor.
5. The positive reactance is a capacitive reactance.
6. Inductive reactance is decreased by increasing frequency.
7. Kirchhoff's Current Law is applied for the nodal analysis at each nodes in circuit.
8. The unit of real power is in W.
9. The more quality factor, the less frequency bandwidth.
10. Resonance occurs when the reactive capacitance equals to reactive resistance.
11. The superposition theorem states that the current and voltage for any element in the circuit is the sum of the currents and voltages produced by each source independently.
12. Equivalent Impedance of Norton and Thevenin equivalent circuit are the same.
13. The primary and secondary currents of a transformer are related by the direct ratios of the primary and secondary turns.
14. The short-circuit equivalent in Thevenin's theorem is a voltage independent source.
15. Ohm's law is applied in the voltage and current sources conversion.
16. In Mesh analysis, the voltage polarity of each R, L and C elements depends on the direction of distinct current of each loop.
17. In Node analysis, 2 nodes provide 2 linear equations.
18. In a Y-connected load, the 220 V is available from line-to-neutral and the 380V is available from line-to-line.
19. In Thevenin's and Norton's theorem, the voltage across the load of both theorems is the same value.
20. The current and voltage of the circuit is in phase, it means that the power factor is unity.
21. The unit of reactive power is VA.
22. The voltage source as shown in Figure below is the dependent voltage source.



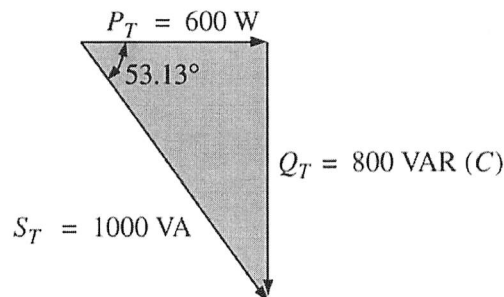
23. Complex conjugate refers to the opposite sign of imaginary part of impedance.
24. High quality factor in series RLC circuit provides narrow bandwidth.
25. In series RLC circuit, frequency resonance depend on resistance.

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Part 2 (40 points) Multiple choice

Instruction: Mark the correct answer for the following questions in the provided answer sheet (see LAST PAGE).

1. What is the power factor of power triangle below ?



- (a) 0.6 (b) 0.8 (c) 0.9 (d) 1.0

2. Convert the current source to a voltage source. Given $I_s = 2\sin(\omega t + 15^\circ)$ A and parallel impedance $Z = 10 + j10$. What is the new voltage source.

- (a) $V_s = 2.8284 \angle 60^\circ$ (b) $V_s = 28.284 \angle 60^\circ$ (c) $V_s = 22.84 \angle 60^\circ$ (d) $V_s = 2.284 \angle 60^\circ$

3. The superposition theorem in AC network analysis, which one of the following statements is true?

- (a) Voltage sources are replaced by open circuits.
 (b) Voltage sources are replaced by short circuits.
 (c) Current sources are replaced by short circuits.
 (d) All impedances are replaced by the complex conjugates.

4. Which one of the following statements is true?

- (a) Voltage sources are replaced by open circuits in Thevenin.
 (b) Current sources are replaced by short circuits in Thevenin.
 (c) Voltage sources are replaced by short circuits in Thevenin.
 (d) All impedances are replaced by the complex conjugates in Thevenin.

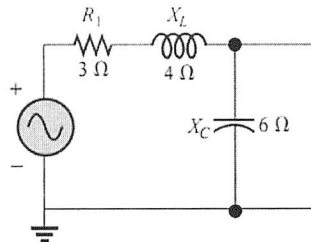
5. If $Z_{th} = 1 - j5 \Omega$, what is the load impedance for maximum power transfer.

- (a) $1 - j5 \Omega$ (b) $1 + j5 \Omega$
 (c) $-10 - j10 \Omega$ (d) $-10 + j10 \Omega$

6. If $S = 5 - j10 \text{ VA}$, which of the following is true?

- (a) Power factor is greater than one.
- (b) Inductive reactance must be greater than capacitive reactance.
- (c) It is pure resistive circuit
- (d) The average power is equal to 5 W

7. From figure below, what is the Thevenin equivalent impedance Z_{TH} at terminal?

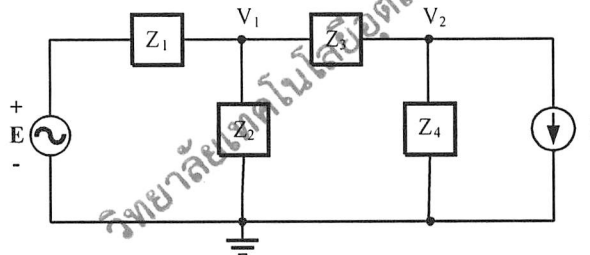


- (a) $(R_1 // jX_L) - jX_C$
- (b) $(R_1 + jX_L) // (-jX_C)$
- (c) $(R_1) // (jX_L) // (-jX_C)$
- (d) $R_1 + jX_L - jX_C$

8. From previous problem, what is the Thevenin equivalent voltage E_{th} ?

- (a) $E / (R_1 + jX_L - jX_C)$
- (b) $E \times (-jX_C) / (R_1 + jX_L - jX_C)$
- (c) $E \times (jX_L - jX_C) / (R_1 + jX_L - jX_C)$
- (d) $E \times R_1 / (R_1 + jX_L - jX_C)$

9. From the circuit below, what is KCL equation at node V_2 ?

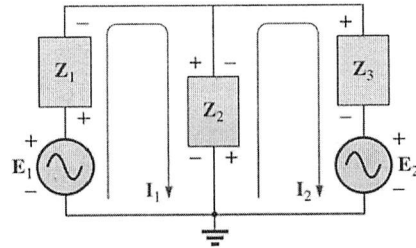


- (a) $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 - V_2)/Z_3 = 0$
- (b) $(V_2 + V_1)/Z_3 + V_2/Z_4 - I = 0$
- (c) $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 + V_2)/Z_3 = 0$
- (d) $(V_2 - V_1)/Z_3 + V_2/Z_4 + I = 0$

10 From the previous problem, what is KCL equation at node V_1 ?

- (a) $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 - V_2)/Z_3 = 0$
- (b) $(V_1 - E)/Z_1 + V_1/Z_2 + (V_1 + V_2)/Z_3 = 0$
- (c) $(V_1 + E)/Z_1 + V_1/Z_2 + (V_1 + V_2)/Z_3 = 0$
- (d) $(V_1 - E)/Z_1 + V_1/Z_2 + (V_1 - V_2)/Z_3 = 0$

11. From the circuit below, what is KVL equation at mesh I_1 ?

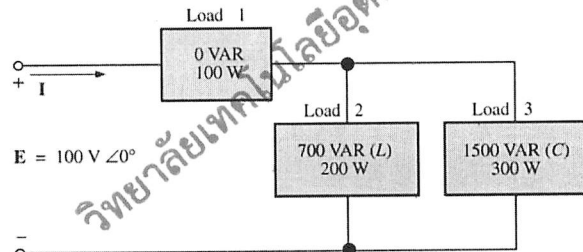


- (a) $E_1 + Z_1 I_1 + Z_2 (I_1 - I_2) = 0$ (b) $E_1 + Z_1 I_1 + Z_2 (I_2 - I_1) = 0$
 (c) $-E_1 + Z_1 I_1 + Z_2 (I_1 - I_2) = 0$ (d) $-E_1 + Z_1 I_1 + Z_2 (I_2 - I_1) = 0$

12. From previous problem, what is KVL equation at mesh I_2 ?

- (a) $E_2 + Z_3 I_2 + Z_2 (I_1 - I_2) = 0$ (b) $E_2 + Z_3 I_2 + Z_2 (I_2 - I_1) = 0$
 (c) $-E_2 + Z_3 I_2 + Z_2 (I_1 - I_2) = 0$ (d) $-E_2 + Z_3 I_2 + Z_2 (I_2 - I_1) = 0$

13. What is the total real power of the system below?



- (a) 600 VA (b) 600 W
 (c) $600 + j2200$ VA (d) $600 - j2200$ VAR

14. What is the reactive power of the previous problem ?

- (a) 800 VAR (b) 800 W
 (c) $600 + j2200$ VA (d) $600 - j2200$ VAR

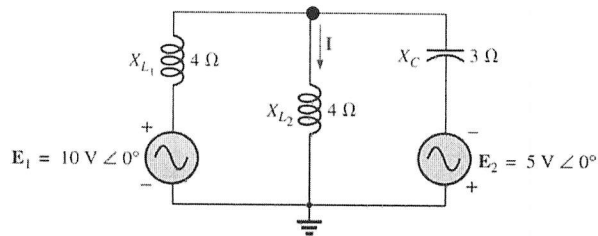
15. What is the apparent power of the previous problem ?

- (a) $600 + j800$ VA (b) $600 - j800$ VA
 (c) $600 + j2200$ VA (d) $600 - j2200$ VA

16. A network consists of $E = 50 \text{ V } \angle 20^\circ$ and $Z = 10 \Omega \angle 30^\circ$ connected in series. If this voltage source is converted to a current source, what is the value of the parallel impedance element Z ?

- (a) $2.0 \Omega \angle 10^\circ$ (b) $0.01 \Omega \angle -30^\circ$
 (c) $10 \Omega \angle 30^\circ$ (d) $0.5 \Omega \angle -10^\circ$

17. Following the superposition theorem, what is the current (I) of 2nd source (E_2)?



- (a) $2.5 \angle 90^\circ$ (b) $2.5 \angle 187.4^\circ$ (c) $2.5 \angle 179.7^\circ$ (d) $2.5 \angle 217.9^\circ$

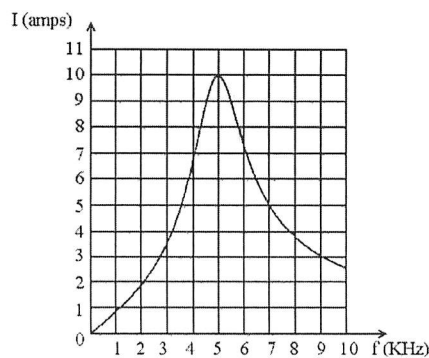
18. From question 17, find the Thevenin equivalent impedance Z_{th} across the terminal at the inductor L_2 .

- (a) $12 \angle -90^\circ \Omega$ (b) $2.5 \angle 90^\circ \Omega$ (c) $2.5 \angle -90^\circ \Omega$ (d) $12 \angle 90^\circ \Omega$

19. From question 17, find the Thevenin equivalent voltage E_{th} across the terminal at the inductor L_2 .

- (a) $E_{th} = 7.69 \angle 0^\circ V$ (b) $E_{th} = 20 \angle 90^\circ V$ (c) $E_{th} = 10 \angle -180^\circ V$ (d) $E_{th} = 7.69 \angle -82.38^\circ V$

20. The curve above describes a series resonant circuit. What is the bandwidth of circuit?

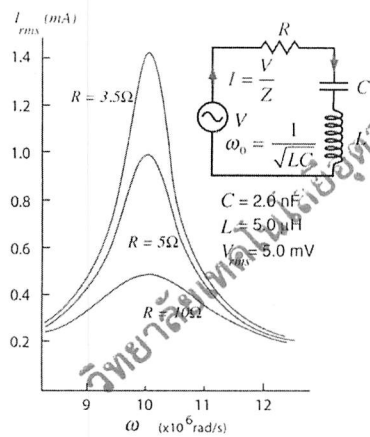


- (a) 1 kHz (b) 2 kHz (c) 3 kHz (d) 4 kHz

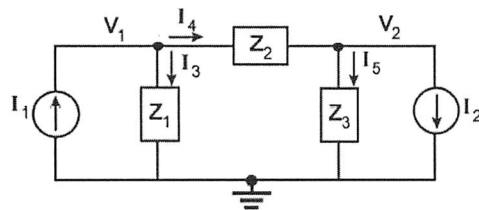
Part 3 (25 points)

Instruction: Show the mathematical expression and answers of following problems.

1. The RLC circuit as shown in Figure below, find resonant frequency (f_s), quality factor (Q_s) and bandwidth (BW)



2. Determine the nodal voltages for the network below



$$Z_1 = 4 \Omega \angle 0^\circ$$

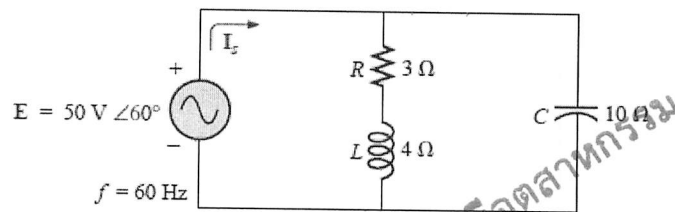
$$Z_2 = 5 \Omega \angle 90^\circ$$

$$Z_3 = 2 \Omega \angle -90^\circ$$

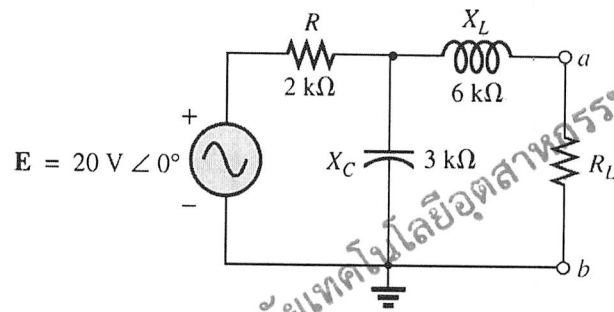
$$I_1 = 3 \text{ A} \angle 0^\circ$$

$$I_2 = 5 \text{ A} \angle 30^\circ$$

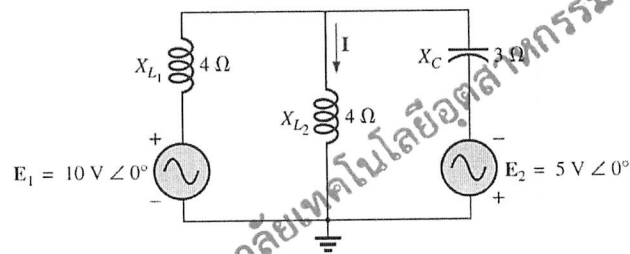
3. From the network in Figure below;
- (a) Find the total number of watts (P), volt-amperes reactive (Q), and volt-amperes (S),
- (b) Sketch the power triangle.



4. From figure below, find the voltage across R_L by using Thevenin theorem.



5. Find voltage across L_2 and current I in circuit in figure below by using Mesh analysis



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